NAMAN SINGH

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PROFESSIONAL SUMMARY

B.Tech Computer Science Engineering (AI & ML) student at VIT Bhopal University (CGPA: 8.42) with corporate Quality Control experience building custom Computer Vision pipelines, deep learning models, and medical image processing solutions. Proficient in Python, OpenCV, PyTorch, CNNs, and Vision Transformers with 109+ LeetCode problems solved; experienced in image segmentation, cell/tissue feature extraction, and REST API deployment. Demonstrates strong scientific problem-solving, quantitative image analysis, and Docker containerization.

TECHNICAL SKILLS

Computer Vision & Image Processing: OpenCV, Image Segmentation, Cell & Tissue Detection, Featurization, Image Enhancement, Denoising, Vision Transformers (ViTs), CNNs (U-Net, MobileNetV2)
Deep Learning & Frameworks: PyTorch, TensorFlow, Scikit-learn, Model Fine-Tuning, Feature Extraction, Quantitative Image Analysis
Programming Languages: Python, C++, Java, SQL, JavaScript, HTML/CSS
Software Architecture & DevOps: REST APIs (FastAPI, Flask), Docker, Redis, AWS (EC2, S3), CI/CD (GitHub Actions), PyTest, Git

PROFESSIONAL EXPERIENCE

Quality Control (QC) Intern | **Aforeserve.com Ltd.** — **May 2026 - Jun 2026**

- Conducted automated image quality audits and computer vision pipeline testing across **5+ core enterprise platforms**, identifying and logging **40+ software defects**.
- Collaborated with software engineering teams to optimize image pre-processing workflows, reducing defect resolution time by **15%**.
- Analyzed diagnostic image processing logs and drafted **3+ quality control reports**, ensuring strict compliance with system accuracy and performance standards.

RELEVANT PROJECTS

Biomedical Microscopy & Cellular Image Segmentation Pipeline
Technologies: Python, OpenCV, PyTorch, U-Net, Vision Transformers (ViTs), NumPy, SciPy

- Engineered a custom deep learning pipeline using OpenCV and PyTorch for cell detection and tissue segmentation across **10,000+ microscopy images** (Fluorescence, Brightfield).
- Implemented U-Net and Vision Transformer (ViT) models to extract morphological features, achieving **95.2% segmentation accuracy** and reducing noise artifacts by **30%**.
- Developed custom featurization algorithms to compute quantitative cell density and spatial intensity metrics with sub-**120ms processing latency**.

Medical Image Analysis & Anomaly Detection System (MRI/CT Scans)
Technologies: Python, PyTorch, OpenCV, MobileNetV2, Flask, Redis, Scikit-learn

- Built an automated medical image analysis engine for volumetric MRI and CT scan anomaly classification, achieving **94.5% diagnostic accuracy**.
- Applied custom image enhancement, CLAHE contrast adjustment, and noise filtering algorithms, improving signal-to-noise ratio by **28%**.
- Deployed a high-throughput Flask REST API integrated with Redis caching to deliver real-time quantitative image evaluations under sub-**100ms response times**.

High-Throughput Quantitative Vision & Feature Extraction Engine
Technologies: Python, OpenCV, TensorFlow, FastAPI, Docker, GitHub Actions, AWS EC2

- Architected an automated computer vision pipeline processing **500,000+ high-resolution biological images** to extract structural and textural features.
- Containerized the vision processing microservices using Docker and automated CI/CD workflows via GitHub Actions, reducing build-integration lead time by **40%**.
- Optimized parallel image matrix computations using OpenCV and NumPy, improving throughput by **35%** on AWS EC2 deployments.

ACHIEVEMENTS & CERTIFICATIONS

- **Harvard CS50's Introduction to Artificial Intelligence with Python:** Completed all **12 of 12 assignments** covering computer vision, neural networks, search algorithms, and transformer models.
- **NPTEL Elite+Gold (Internet of Things - IIT Kharagpur):** Ranked in the **Top 5% of 10,913+ candidates** with a score of **93%**, mastering sensor data processing, imaging hardware, and software-hardware integration.
- **NPTEL Elite+Silver (Cloud Computing - IIT Kharagpur):** Certified in distributed cloud architecture across **4** virtualized test deployments.

LEADERSHIP & VOLUNTEERING

- **IEEE CSNT-2025 Student Coordinator (VIT Bhopal):** Co-coordinated international technology conference hosting **500+ attendees**; led **3 technical presentation tracks** and managed event logistics.

EDUCATION

VIT Bhopal University — **Sep 2023 - Jun 2027 (Expected)**
B.Tech, Computer Science Engineering (AI & ML Specialization)
CGPA: 8.42 / 10.0 | Coursework: Data Structures & Algorithms, Deep Learning, NLP, Database Management

Holy Cross School, Bengaluru — **Jun 2021 - Mar 2023**
Secondary & Senior Secondary Education
Class X: **95%** | Class XII: **85%**